Lecture 9: Trees, Cutsets and Fundamental Loops

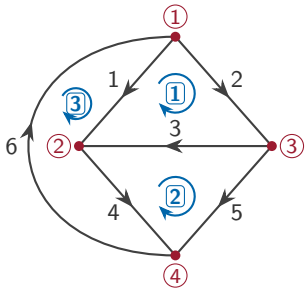
EE201 – Circuit Theory I

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current voltage, cutsets, loops tree nodes

Review: two matrices for one graph



$n = 4, \quad b = 6$

Incidence matrix (rows: nodes ①, ②, ③)

$$A = \begin{array}{c} \\ ① \\ ② \\ ③ \end{array} \begin{array}{c} \begin{array}{cccccc} 1 & 2 & 3 & 4 & 5 & 6 \end{array} \\ \begin{bmatrix} 1 & 1 & 0 & 0 & 0 & -1 \\ -1 & 0 & -1 & 1 & 0 & 0 \\ 0 & -1 & 1 & 0 & 1 & 0 \end{bmatrix} \end{array}$$

Mesh matrix (rows: meshes 1, 2, 3)

$$M = \begin{array}{c} \\ 1 \\ 2 \\ 3 \end{array} \begin{array}{c} \begin{array}{cccccc} 1 & 2 & 3 & 4 & 5 & 6 \end{array} \\ \begin{bmatrix} -1 & 1 & 1 & 0 & 0 & 0 \\ 0 & 0 & -1 & -1 & 1 & 0 \\ 1 & 0 & 0 & 1 & 0 & 1 \end{bmatrix} \end{array}$$

$n - 1 = 3$ independent KCL equations, $\quad b - n + 1 = 3$ independent KVL equations

Every row of A is orthogonal to every row of M: $\quad AM^T = \mathbf{0}$

Review: node voltages and mesh currents

	node voltages	**mesh currents**
works for	any connected graph	planar, nonseparable
number of unknowns	$n-1$	$b-n+1$
holds automatically	KVL: $\mathbf{v} = A^T\mathbf{e}$	KCL: $\mathbf{i} = M^T\mathbf{i}_M$
left to write	KCL: $A\mathbf{i} = \mathbf{0}$	KVL: $M\mathbf{v} = \mathbf{0}$

Tellegen: $\mathbf{i}^T\mathbf{v} = 0$

Why this lecture?

- **Goal:** solve a circuit, i.e. find all b branch currents and b branch voltages.

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- **So far:** node voltages with A work for any graph; mesh currents with M only for planar, nonseparable graphs.

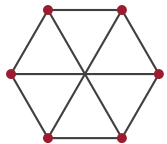
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- **Missing:** a mesh-like tool for every connected graph.

Why this lecture?

- **Goal:** solve a circuit, i.e. find all b branch currents and b branch voltages.
- **We need:** enough independent KCL and KVL equations, and as few unknowns as possible.
- **So far:** node voltages with A work for any graph; mesh currents with M only for planar, nonseparable graphs.
- **Missing:** a mesh-like tool for every connected graph.
- **Today:** pick a tree. It gives the KCL and KVL equations, and the unknowns, for any connected graph.

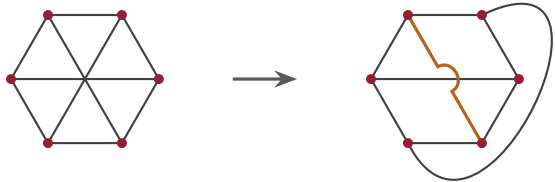
Nonplanar circuits



crossings at the centre

Can this graph be redrawn without crossings?

Nonplanar circuits

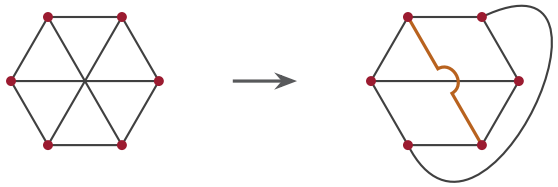


crossings at the centre

jumper wire: still one hop

No: at least one hop remains $\Rightarrow$ no meshes, no mesh matrix.

Nonplanar circuits



crossings at the centre

jumper wire: still one hop

No: at least one hop remains ⇒ no meshes, no mesh matrix.

Question

Without meshes, how do we find $n-1$ independent KCL and $b-n+1$ independent KVL equations?

Today: a method that needs no planar drawing and works for any connected graph.

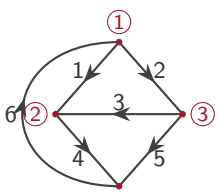
Roadmap

1. **Trees** — a skeleton that reaches every node, with no loop
2. **Cutsets** — KCL on closed surfaces, not only on single nodes
3. **Fundamental cutsets** — a tree picks $n-1$ independent KCL equations (matrix Q)
4. **Fundamental loops** — the same tree picks $b-n+1$ independent KVL equations (matrix B)
5. **Any connected graph** — nonplanar and separable graphs too

Trees

Tree

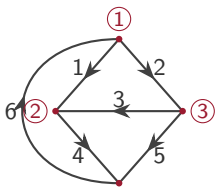
A subgraph T of a connected graph G is a tree if it contains all the nodes of G, is connected, and contains no loops.



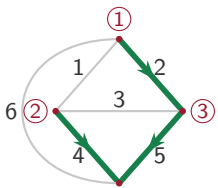
Trees

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graph G

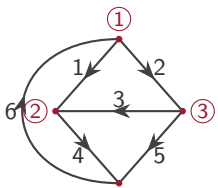


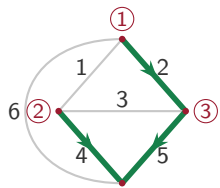
$\{2, 4, 5\}$: **tree?**

Trees

Tree

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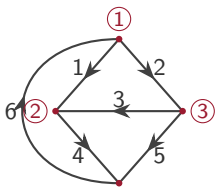




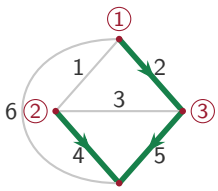
Trees

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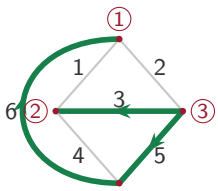
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$\{2, 4, 5\}$: **tree**

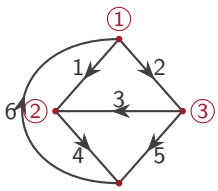


$\{3, 5, 6\}$: **tree?**

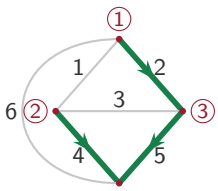
Trees

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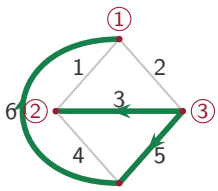
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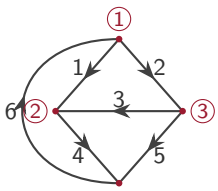


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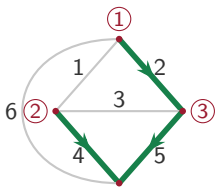
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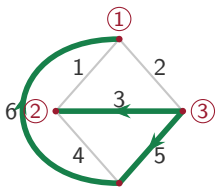
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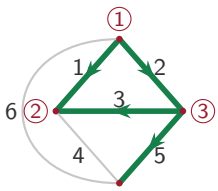
graph G



$\{2, 4, 5\}$: **tree**



$\{3, 5, 6\}$: **tree**

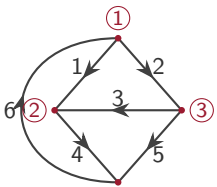


$\{1, 2, 3, 5\}$: **tree?**

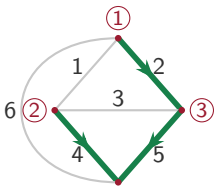
Trees

Tree

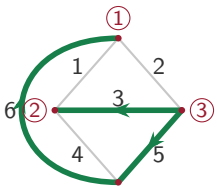
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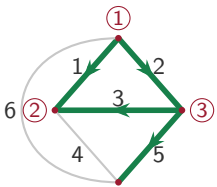
graph G



$\{2, 4, 5\}$: **tree**



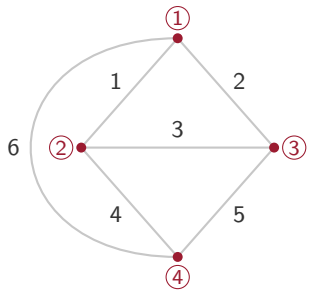
$\{3, 5, 6\}$: **tree**



$\{1, 2, 3, 5\}$: **not a tree**

Branches 1, 2, 3 form a loop. A graph has many trees.

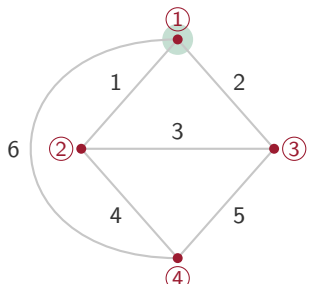
How many branches does a tree have?



$n = 4$ nodes, $b = 6$ branches

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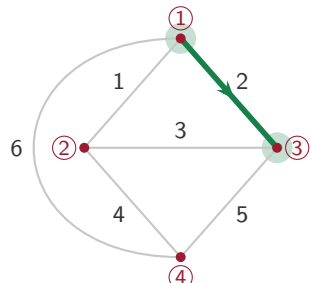
- Start from one node; add tree branches one at a time.



start at node ①

How many branches does a tree have?

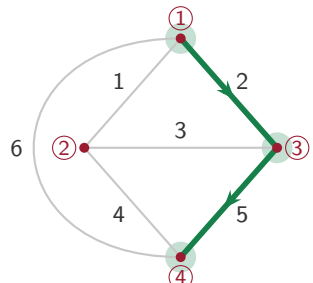
- Start from one node; add tree branches one at a time.
- Each new branch must reach a new node — otherwise it closes a loop.



branch 2 reaches node ③

How many branches does a tree have?

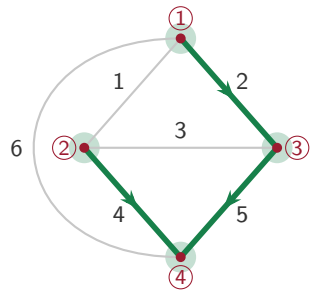
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branch 5 reaches node ④

How many branches does a tree have?

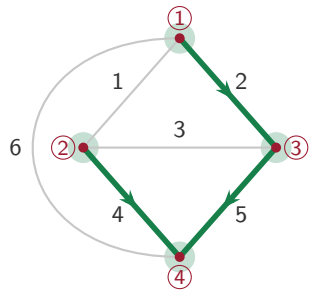
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branch 4 reaches node ②

How many branches does a tree have?

- Start from one node; add tree branches one at a time.
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- Each branch adds one node. All n nodes are reached after $n-1$ branches.

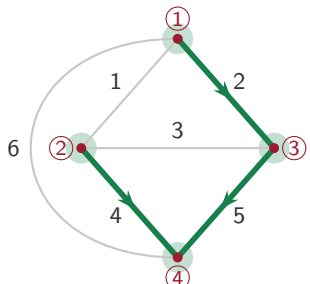


4 nodes, 3 tree branches: $n-1=3$

How many branches does a tree have?

- Start from one node; add tree branches one at a time.
- Each new branch must reach a new node — otherwise it closes a loop.
- Each branch adds one node. All n nodes are reached after $n-1$ branches.

The other $b-n+1$ branches are left out.



left out: 1, 3, 6 $(b-n+1=3)$

Roadmap

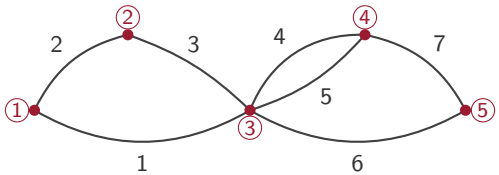
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2. **Cutsets** — KCL on closed surfaces, not only on single nodes
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5. **Any connected graph** — nonplanar and separable graphs too

Cutsets

Cutset

A set C of branches of a connected graph is a cutset if

1. removing all of C leaves the graph unconnected, and
2. removing all of C except any one of them leaves it connected.

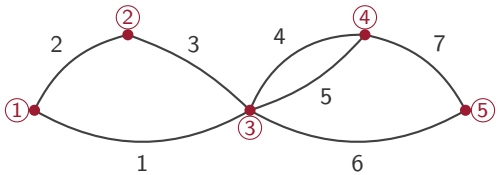


Cutsets

Cutset

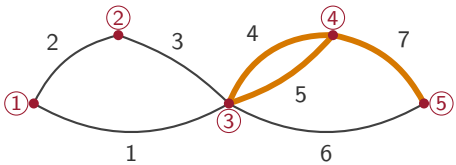
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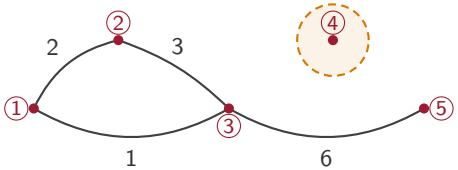
In words: removing the set cuts the graph into two pieces, and every branch in the set is needed for that.

Example: cutsets



$$C_1 = \{4, 5, 7\}$$

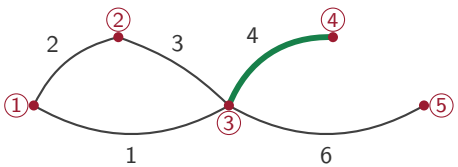
Example: cutsets



$C_1 = \{4, 5, 7\}$

(1) Remove all three: node ④ is cut off. ✓

Example: cutsets



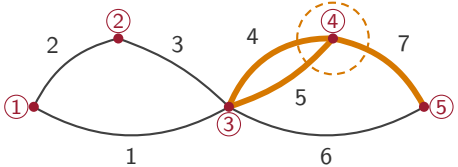
$C_1 = \{4, 5, 7\}$

(1) Remove all three: node ④ is cut off. ✓

(2) Put back one, e.g. 4: node ④ is reached again. Same for 5 or 7. ✓

$\Rightarrow$ C_1 is a cutset.

Example: cutsets



$C_1 = \{4, 5, 7\}$

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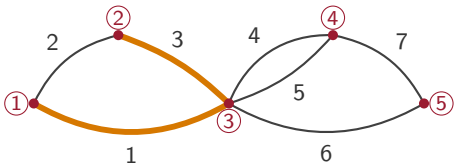
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$\Rightarrow$ C_1 is a cutset.

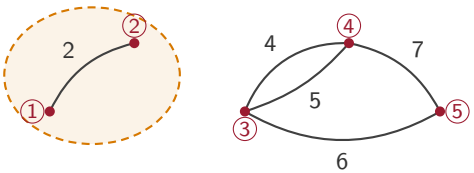
All branches back: the closed surface around node ④ crosses exactly 4, 5, 7.

Example: cutsets

$$C_2 = \{1, 3\}$$



Example: cutsets



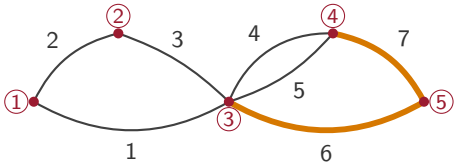
$C_2 = \{1, 3\}$

(1) Remove both: nodes ①, ② are cut off. ✓

(2) Put back 1 or 3: connected again. ✓

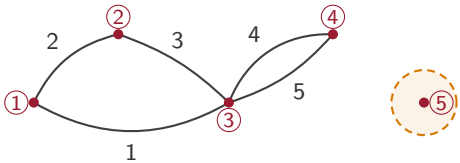
$\Rightarrow$ C_2 is a cutset.

Example: cutsets



$C_3 = \{6, 7\}$

Example: cutsets



$C_3 = \{6, 7\}$

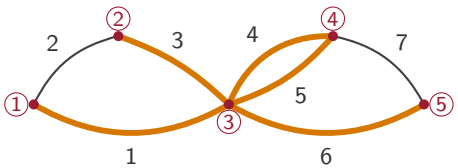
(1) Remove both: node ⑤ is cut off. ✓

(2) Put back 6 or 7: connected again. ✓

$\Rightarrow$ C_3 is a cutset.

Cutsets can have different sizes: C_1 has 3 branches, C_2 and C_3 have 2.

Example: not every set is a cutset



$C = \{1, 3, 4, 5, 6\}$: a cutset?

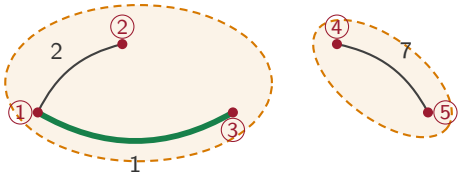
Example: not every set is a cutset



$C = \{1, 3, 4, 5, 6\}$: a cutset?

(1) Remove all five: three pieces. ✓

Example: not every set is a cutset



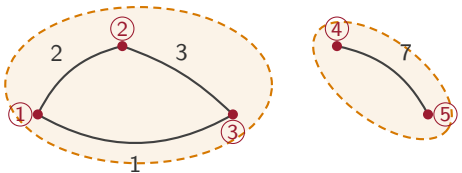
$C = \{1, 3, 4, 5, 6\}$: a cutset?

(1) Remove all five: three pieces. ✓

(2) Put back 1: still unconnected. ×

$\Rightarrow$ not a cutset: some branches are not needed.

Example: not every set is a cutset



$C = \{1, 3, 4, 5, 6\}$: a cutset?

(1) Remove all five: three pieces. ✓

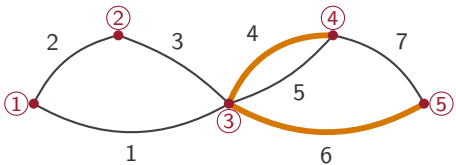
(2) Put back 1: still unconnected. ×

$\Rightarrow$ not a cutset: some branches are not needed.

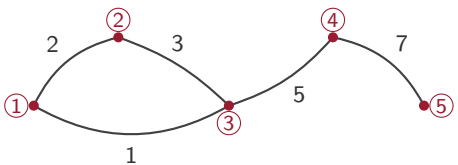
Branches 1, 3 are extra: $\{4, 5, 6\}$ alone is a cutset.

Example: not every set is a cutset

$C = \{4, 6\}$: a cutset?



Example: not every set is a cutset

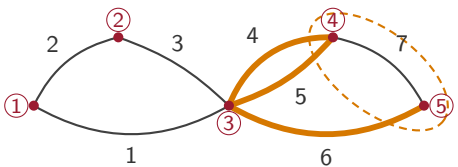


$C = \{4, 6\}$: a cutset?

(1) Remove both: still one piece — branch 5 still joins ③ and ④. ×

$\Rightarrow$ not a cutset: a branch is missing.

Example: not every set is a cutset



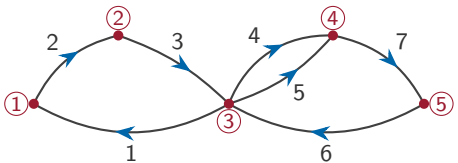
$C = \{4, 6\}$: a cutset?

(1) Remove both: still one piece — branch 5 still joins ③ and ④. ×

$\Rightarrow$ not a cutset: a branch is missing.

Branches 4 and 5 both join ③ and ④ (parallel): a surface that crosses one crosses the other. Here the surface gives $\{4, 5, 6\}$.

KCL on a cutset



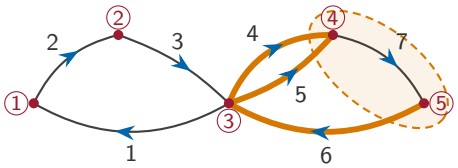
branch	1	2	3	4	5	6	7
current (A)	2	2	2	1	2	3	3

Branch currents in the arrow directions.

They satisfy KCL at every node, e.g. node ③:

$$i_1 + i_4 + i_5 - i_3 - i_6 = 2+1+2-2-3 = 0.$$

KCL on a cutset



branch	1	2	3	4	5	6	7
current (A)	2	2	2	1	2	3	3

Cutset $\{4, 5, 6\}$: surface around ④, ⑤

leaving: i_6; entering: i_4, i_5

$$i_6 - i_4 - i_5 = 3 - 1 - 2 = 0 \checkmark$$

KCL at ④: $i_7 - i_4 - i_5 = 0$
KCL at ⑤: $i_6 - i_7 = 0$

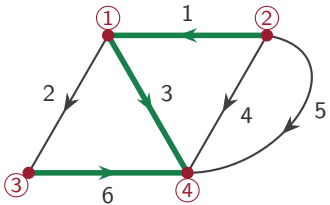
Add them: i_7 is inside the surface and cancels.

KCL on a cutset: the cutset currents add up to zero
(+ leaving the surface, − entering).

Roadmap

1. **Trees** — a skeleton that reaches every node, with no loop ✓
2. **Cutsets** — KCL on closed surfaces, not only on single nodes ✓
3. **Fundamental cutsets** — a tree picks $n-1$ independent KCL equations (matrix Q)
4. **Fundamental loops** — the same tree picks $b-n+1$ independent KVL equations (matrix B)
5. **Any connected graph** — nonplanar and separable graphs too

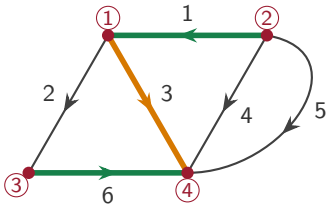
Fundamental cutsets



tree $\{1, 3, 6\}$, co-tree $\{2, 4, 5\}$

1. Fix a tree (green). The other branches form the co-tree.

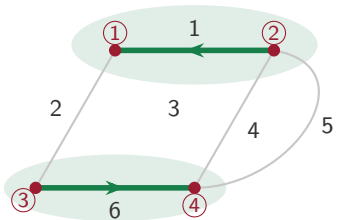
Fundamental cutsets



tree $\{1, 3, 6\}$, co-tree $\{2, 4, 5\}$

1. Fix a tree (green). The other branches form the co-tree.
2. Pick one tree branch, e.g. 3.

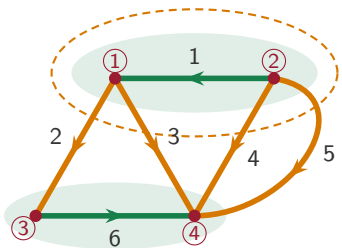
Fundamental cutsets



tree $\{1, 3, 6\}$, co-tree $\{2, 4, 5\}$

1. Fix a tree (green). The other branches form the co-tree.
2. Pick one tree branch, e.g. 3.
3. Remove it: the tree falls into two parts, $\{①, ②\}$ and $\{③, ④\}$.

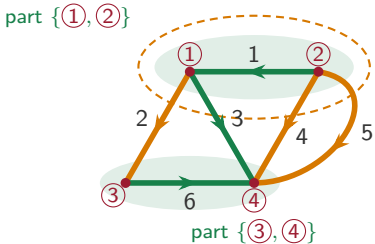
Fundamental cutsets



tree $\{1, 3, 6\}$, co-tree $\{2, 4, 5\}$

1. Fix a tree (green). The other branches form the co-tree.
2. Pick one tree branch, e.g. 3.
3. Remove it: the tree falls into two parts, $\{①, ②\}$ and $\{③, ④\}$.
4. The branches joining the two parts form a cutset: $\{3, 2, 4, 5\}$. It is the fundamental cutset of 3.

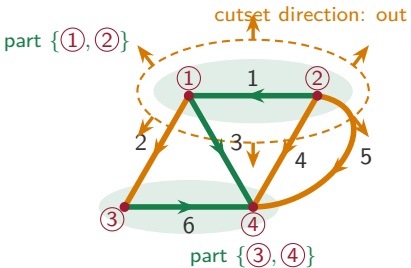
Fundamental cutset: one tree branch, one direction



Cutset of 3: $\{3, 2, 4, 5\}$

One tree branch. Only 3 is a tree branch; 2, 4, 5 are co-tree branches. The other tree branches (1, 6) lie inside one part.

Fundamental cutset: one tree branch, one direction

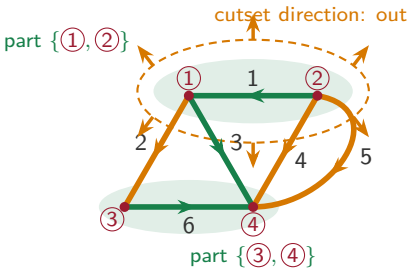


Cutset of 3: $\{3, 2, 4, 5\}$

One tree branch. Only 3 is a tree branch; 2, 4, 5 are co-tree branches. The other tree branches (1, 6) lie inside one part.

Direction. Branch 3 leaves the surface, from part $\{①, ②\}$ to part $\{③, ④\}$: the cutset points out.

Fundamental cutset: one tree branch, one direction



Cutset of 3: $\{3, 2, 4, 5\}$

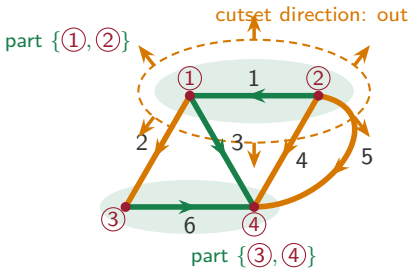
One tree branch. Only 3 is a tree branch; 2, 4, 5 are co-tree branches. The other tree branches (1, 6) lie inside one part.

Direction. Branch 3 leaves the surface, from part $\{①, ②\}$ to part $\{③, ④\}$: the cutset points out.

KCL. Branch k carries current i_k along its arrow. KCL on the surface: $+$ leaving, $-$ entering. Branches 2, 4, 5 leave too:

$$i_3 + i_2 + i_4 + i_5 = 0.$$

Fundamental cutset: one tree branch, one direction



Fundamental cutset

- the tree branch + the co-tree branches joining the two parts
- only one tree branch in it
- direction: that of the tree branch

Cutset of 3: $\{3, 2, 4, 5\}$

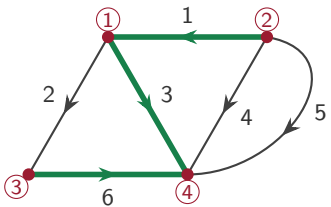
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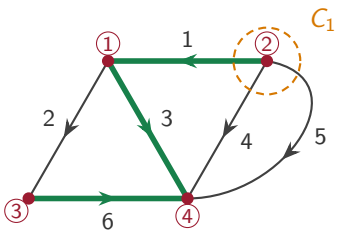
Example: fundamental cutsets of G



For each tree branch: its fundamental cutset, and KCL on that cutset (branch currents i_k along the arrows; $+$ if a branch crosses the surface the same way as the tree branch, $-$ if not).

tree br.	cutset	KCL on it

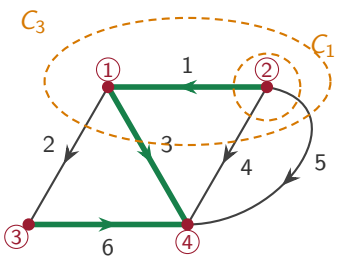
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tree br.	cutset	KCL on it
1	$\{1, 4, 5\}$	$i_1 + i_4 + i_5 = 0$

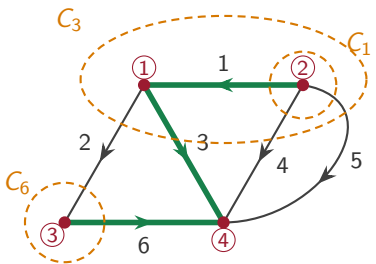
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tree br.	cutset	KCL on it
1	$\{1, 4, 5\}$	$i_1 + i_4 + i_5 = 0$
3	$\{3, 2, 4, 5\}$	$i_3 + i_2 + i_4 + i_5 = 0$

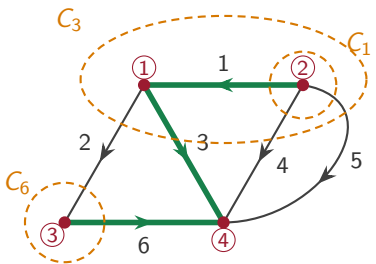
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tree br.	cutset	KCL on it
1	$\{1, 4, 5\}$	$i_1 + i_4 + i_5 = 0$
3	$\{3, 2, 4, 5\}$	$i_3 + i_2 + i_4 + i_5 = 0$
6	$\{6, 2\}$	$i_6 - i_2 = 0$

Example: fundamental cutsets of G

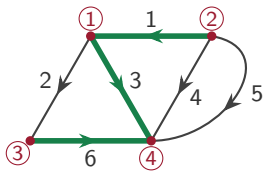


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3	$\{3, 2, 4, 5\}$	$i_3 + i_2 + i_4 + i_5 = 0$
6	$\{6, 2\}$	$i_6 - i_2 = 0$

$n - 1 = 3$ KCL equations, independent: each has its own tree branch current.

Matrix form: fundamental cutset matrix Q



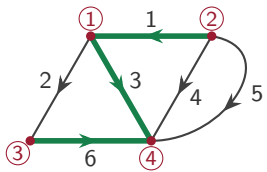
tree $\{1, 3, 6\}$

Collect the fundamental cutset KCL equations in a matrix:

one row per tree branch, one column per branch;
$+1$ / -1: branch in the cutset, same / opposite direction;
0: not in it.

$$Q = \begin{array}{c} \\ 1 \\ 3 \\ 6 \end{array} \begin{array}{c} \begin{array}{cccccc} 1 & 2 & 3 & 4 & 5 & 6 \end{array} \\ \begin{bmatrix} 1 & 0 & 0 & 1 & 1 & 0 \\ 0 & 1 & 1 & 1 & 1 & 0 \\ 0 & -1 & 0 & 0 & 0 & 1 \end{bmatrix} \end{array}$$

Matrix form: fundamental cutset matrix Q



tree $\{1, 3, 6\}$

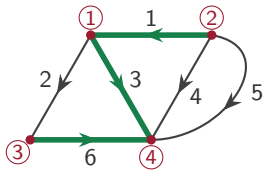
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KCL: $Q\mathbf{i} = \mathbf{0}$ (like $A\mathbf{i} = \mathbf{0}$)

Matrix form: fundamental cutset matrix Q



tree $\{1, 3, 6\}$

Collect the fundamental cutset KCL equations in a matrix:

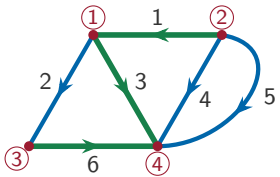
one row per tree branch, one column per branch;
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KCL: $Q\mathbf{i} = \mathbf{0}$ (like $A\mathbf{i} = \mathbf{0}$)

Tree columns 1, 3, 6 form an identity: each row has its own tree branch, so the rows are independent.

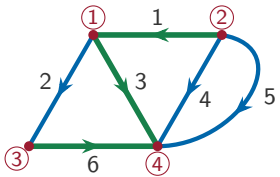
Branch currents from co-tree currents



tree $\{1, 3, 6\}$, co-tree $\{2, 4, 5\}$

Goal. Relate every branch current to the co-tree currents i_2, i_4, i_5.

Branch currents from co-tree currents

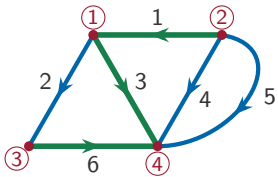


tree $\{1, 3, 6\}$, co-tree $\{2, 4, 5\}$

Goal. Relate every branch current to the co-tree currents i_2, i_4, i_5.

Co-tree branches: nothing to do — i_2, i_4, i_5 themselves.

Branch currents from co-tree currents



tree $\{1, 3, 6\}$, co-tree $\{2, 4, 5\}$

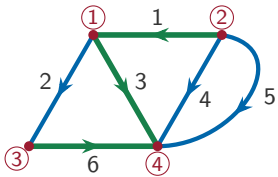
Goal. Relate every branch current to the co-tree currents i_2, i_4, i_5.

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Tree branches: each fundamental cutset has one tree branch. Solve its KCL for that current:

$$i_1 = -i_4 - i_5, \quad i_3 = -i_2 - i_4 - i_5, \quad i_6 = i_2$$

Branch currents from co-tree currents



tree $\{1, 3, 6\}$, co-tree $\{2, 4, 5\}$

Goal. Relate every branch current to the co-tree currents i_2, i_4, i_5.

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$$i_1 = -i_4 - i_5, \quad i_3 = -i_2 - i_4 - i_5, \quad i_6 = i_2$$

So all b branch currents are combinations of the $b - n + 1$ co-tree currents, and KCL holds automatically — like mesh currents.

In matrix form: $\mathbf{i} = B^T \mathbf{i}_C$

Every branch current from the co-tree currents $\mathbf{i}_C = [i_2 \ i_4 \ i_5]^T$:

$$\begin{bmatrix} i_1 \\ i_2 \\ i_3 \\ i_4 \\ i_5 \\ i_6 \end{bmatrix} = \begin{bmatrix} 0 & -1 & -1 \\ 1 & 0 & 0 \\ -1 & -1 & -1 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} i_2 \\ i_4 \\ i_5 \end{bmatrix} \quad \begin{array}{l} \leftarrow i_1 = -i_4 - i_5 \\ \\ \leftarrow i_3 = -i_2 - i_4 - i_5 \\ \\ \\ \leftarrow i_6 = i_2 \end{array}$$

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We call this 6×3 matrix B^T: $\mathbf{i} = B^T \mathbf{i}_C$, like $\mathbf{i} = M^T \mathbf{i}_M$. KCL holds for any $\mathbf{i}_C$.

In matrix form: $\mathbf{i} = B^T \mathbf{i}_C$

Every branch current from the co-tree currents $\mathbf{i}_C = [i_2 \ i_4 \ i_5]^T$:

$$\begin{bmatrix} i_1 \\ i_2 \\ i_3 \\ i_4 \\ i_5 \\ i_6 \end{bmatrix} = \begin{bmatrix} 0 & -1 & -1 \\ 1 & 0 & 0 \\ -1 & -1 & -1 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} i_2 \\ i_4 \\ i_5 \end{bmatrix} \quad \begin{array}{l} \leftarrow i_1 = -i_4 - i_5 \\ \\ \leftarrow i_3 = -i_2 - i_4 - i_5 \\ \\ \\ \leftarrow i_6 = i_2 \end{array}$$

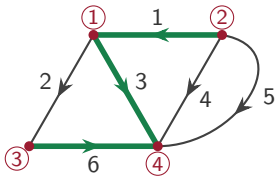
We call this 6×3 matrix B^T: $\mathbf{i} = B^T \mathbf{i}_C$, like $\mathbf{i} = M^T \mathbf{i}_M$. KCL holds for any $\mathbf{i}_C$.

Its transpose B (rows: co-tree branches 2, 4, 5) will appear again with the fundamental loops.

Roadmap

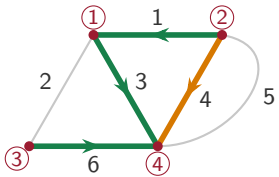
1. **Trees** — a skeleton that reaches every node, with no loop ✓
2. **Cutsets** — KCL on closed surfaces, not only on single nodes ✓
3. **Fundamental cutsets** — a tree picks $n-1$ independent KCL equations (matrix Q) ✓
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5. **Any connected graph** — nonplanar and separable graphs too

Fundamental loops



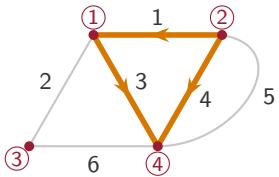
1. Fix a tree (green): $\{1, 3, 6\}$. Co-tree: $\{2, 4, 5\}$.

Fundamental loops



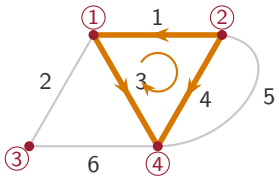
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Fundamental loops



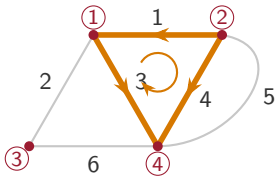
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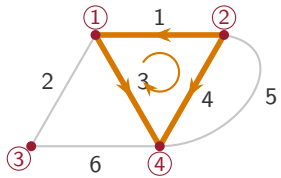
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$$v_4 - v_3 - v_1 = 0.$$

Fundamental loops



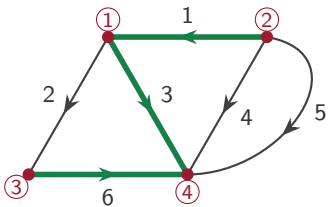
Fundamental loop

- the co-tree branch + the tree path between its ends
- only one co-tree branch in it
- direction: that of the co-tree branch

1. Fix a tree (green): $\{1, 3, 6\}$. Co-tree: $\{2, 4, 5\}$.
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Example: fundamental loops of G

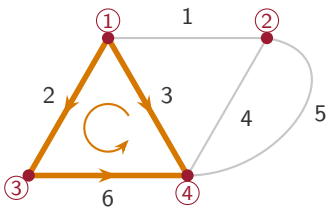


tree $\{1, 3, 6\}$, co-tree $\{2, 4, 5\}$

For each co-tree branch: its fundamental loop, and KVL around it (walk in the direction of the co-tree branch; $+v_k$ along an arrow, $-v_k$ against).

co-tree br.	loop	KVL around it

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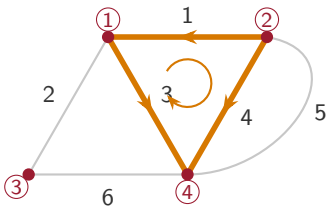


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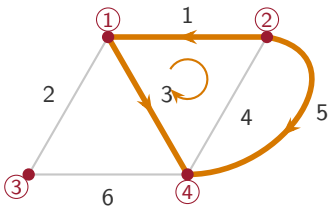


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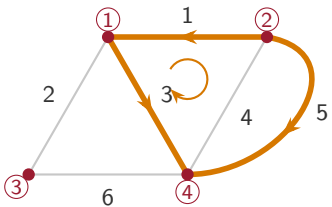


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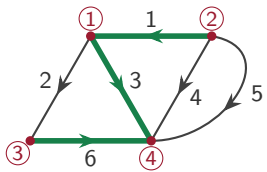
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$b - n + 1 = 3$ KVL equations, independent: each has its own co-tree branch voltage.

Matrix form: fundamental loop matrix B



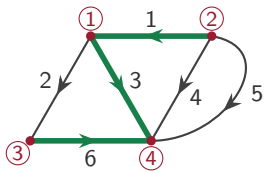
co-tree $\{2, 4, 5\}$

Collect the fundamental loop KVL equations in a matrix:

one row per co-tree branch, one column per branch;
$+1$ / -1: branch in the loop, along / against its direction;
0: not in it.

$$B = \begin{array}{c} \\ 2 \\ 4 \\ 5 \end{array} \begin{array}{c} \begin{array}{cccccc} 1 & 2 & 3 & 4 & 5 & 6 \end{array} \\ \begin{bmatrix} 0 & 1 & -1 & 0 & 0 & 1 \\ -1 & 0 & -1 & 1 & 0 & 0 \\ -1 & 0 & -1 & 0 & 1 & 0 \end{bmatrix} \end{array}$$

Matrix form: fundamental loop matrix B



co-tree $\{2, 4, 5\}$

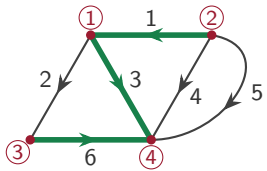
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KVL: $B\mathbf{v} = \mathbf{0}$ (like $M\mathbf{v} = \mathbf{0}$)

Matrix form: fundamental loop matrix B



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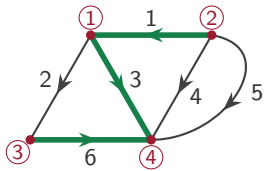
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Matrix form: fundamental loop matrix B



co-tree $\{2, 4, 5\}$

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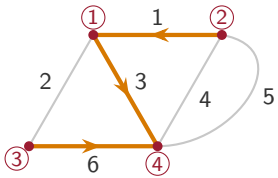
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Co-tree columns 2, 4, 5 form an identity: each row has its own co-tree branch, so the rows are independent.
Compare with $\mathbf{i} = B^T \mathbf{i}_C$: the same matrix B.

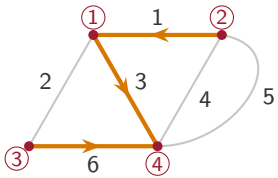
Branch voltages from tree voltages



tree $\{1, 3, 6\}$, co-tree $\{2, 4, 5\}$

Goal. Relate every branch voltage to the tree voltages v_1, v_3, v_6.

Branch voltages from tree voltages

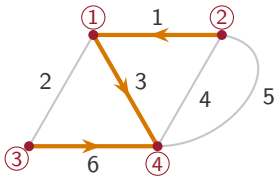


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Branch voltages from tree voltages



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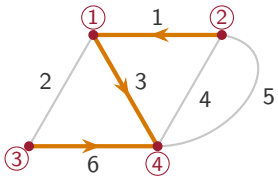
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Co-tree branches: each fundamental loop has one co-tree branch. Solve its KVL for that voltage:

$$v_2 = v_3 - v_6, \quad v_4 = v_1 + v_3, \quad v_5 = v_1 + v_3$$

Branch voltages from tree voltages



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So all b branch voltages are combinations of the $n - 1$ tree voltages, and KVL holds automatically — like node voltages.

In matrix form: $\mathbf{v} = Q^T \mathbf{v}_T$

Every branch voltage from the tree voltages $\mathbf{v}_T = [v_1 \; v_3 \; v_6]^T$:

$$\begin{bmatrix} v_1 \\ v_2 \\ v_3 \\ v_4 \\ v_5 \\ v_6 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & -1 \\ 0 & 1 & 0 \\ 1 & 1 & 0 \\ 1 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} v_1 \\ v_3 \\ v_6 \end{bmatrix} \begin{array}{l} \\ \leftarrow v_2 = v_3 - v_6 \\ \\ \leftarrow v_4 = v_1 + v_3 \\ \leftarrow v_5 = v_1 + v_3 \\ \\ \end{array}$$

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And the two matrices fit together: $QB^T = \mathbf{0}$, like $AM^T = \mathbf{0}$.

Roadmap

1. **Trees** — a skeleton that reaches every node, with no loop ✓
2. **Cutsets** — KCL on closed surfaces, not only on single nodes ✓
3. **Fundamental cutsets** — a tree picks $n-1$ independent KCL equations (matrix Q) ✓
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5. **Any connected graph** — nonplanar and separable graphs too

Any connected graph

	how many	gives
tree branches	$n-1$	fundamental cutsets: the KCL equations
co-tree branches	$b-n+1$	fundamental loops: the KVL equations

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No drawing needed: works for nonplanar graphs.

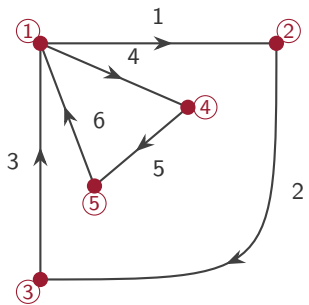
Any connected graph

	how many	gives
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It also works for separable graphs. Example next.

Example: the separable graph

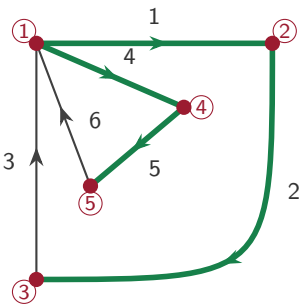


$n = 5, \quad b = 6$

Two loops joined only at node ①.

Separable: the mesh method did not work here.

Example: the separable graph

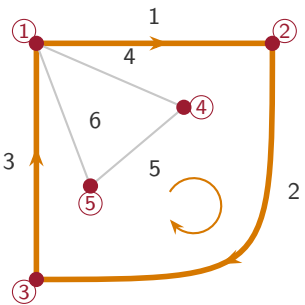


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Tree $\{1, 2, 4, 5\}$: $n - 1 = 4$ branches.

Co-tree $\{3, 6\}$: $b - n + 1 = 2$ branches.

Example: the separable graph



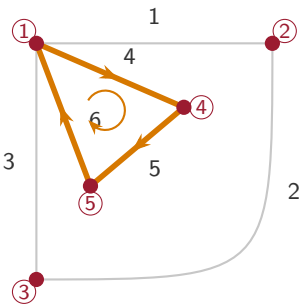
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Fundamental loops: KVL around each, walking with the co-tree branch

co-tree br.	loop	KVL
3	$\{3, 1, 2\}$	$v_3 + v_1 + v_2 = 0$

The loop of 3 is the outer loop.

Example: the separable graph



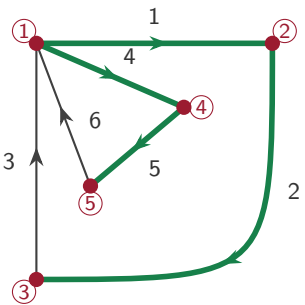
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3	$\{3, 1, 2\}$	$v_3 + v_1 + v_2 = 0$
6	$\{6, 4, 5\}$	$v_6 + v_4 + v_5 = 0$

The loop of 6 is the inner triangle.

Example: the separable graph

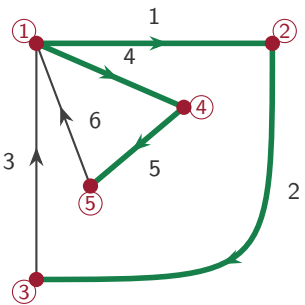


$n = 5, \quad b = 6$

Fundamental cutsets: KCL on each, signs set by the tree branch

tree br.	cutset	KCL
1	$\{1,3\}$	$i_1 - i_3 = 0$
2	$\{2,3\}$	$i_2 - i_3 = 0$
4	$\{4,6\}$	$i_4 - i_6 = 0$
5	$\{5,6\}$	$i_5 - i_6 = 0$

Example: the separable graph



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4	$\{4,6\}$	$i_4 - i_6 = 0$
5	$\{5,6\}$	$i_5 - i_6 = 0$

So $i_1 = i_2 = i_3$ and $i_4 = i_5 = i_6$:
one current for each loop.

4 KCL + 2 KVL equations, all independent — no problem at node ①.

Looking ahead

	incidence / mesh	**fundamental cutset / loop**
KCL	$A\mathbf{i} = \mathbf{0}$	$Q\mathbf{i} = \mathbf{0}$
KVL	$M\mathbf{v} = \mathbf{0}$	$B\mathbf{v} = \mathbf{0}$
$n-1$ voltages	node voltages: $\mathbf{v} = A^T\mathbf{e}$	tree voltages: $\mathbf{v} = Q^T\mathbf{v}_T$
$b-n+1$ currents	mesh currents: $\mathbf{i} = M^T\mathbf{i}_M$	co-tree currents: $\mathbf{i} = B^T\mathbf{i}_C$
works for	A: any; M: planar, nonseparable	any connected graph

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These unknowns are what circuit analysis will solve for:
far fewer than the $2b$ branch currents and voltages.

Summary

- **Tree**: all nodes, connected, no loop; $n-1$ branches. The rest: the **co-tree**, $b-n+1$ branches.
- **Cutset**: minimal set of branches that cuts the graph in two. KCL holds on it.
- **Fundamental cutsets** (one per tree branch): $n-1$ independent KCL equations. Every branch current follows from the co-tree currents.
- **Fundamental loops** (one per co-tree branch): $b-n+1$ independent KVL equations. Every branch voltage follows from the tree voltages.
- Matrix form: $Q\mathbf{i} = \mathbf{0}$, $B\mathbf{v} = \mathbf{0}$, $\mathbf{i} = B^T\mathbf{i}_C$, $\mathbf{v} = Q^T\mathbf{v}_T$.
- Works for any connected graph — planar or not, separable or not.

Questions?